

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

**CSA1511 – CLOUD COMPUTING AND BIG DATA
ANALYTICS**

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Application Based

1. Auto-Scaling and Load Balancing for Seasonal E-Commerce Traffic

Problem Statement

An e-commerce company experiences unpredictable traffic spikes during seasonal sales. During events such as Diwali sales, Black Friday, or flash sales, thousands or millions of users may access the website simultaneously.

Application of Cloud Auto-Scaling

Cloud auto-scaling automatically adjusts the number of application servers according to demand.

Normal Traffic:

Users → Load Balancer → Few Application Servers

Sale Period:

Users → Load Balancer → Many Application Servers

When traffic increases, auto-scaling launches additional instances.

When traffic decreases, unnecessary instances are removed.

Role of Load Balancing

A load balancer distributes incoming requests across multiple servers.

For example:

10,000 users

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Load Balancer



Server 1 | Server 2 | Server 3

This prevents a single server from becoming overloaded.

How It Reduces Downtime

- Prevents server overload.
- Automatically adds capacity during traffic spikes.
- Distributes requests across healthy servers.
- Removes failed servers from traffic.
- Maintains application availability.

Improving User Experience

Users experience:

- Faster page loading.
- Fewer failed requests.
- Reduced waiting time.
- Better checkout performance.
- Fewer website crashes during sales.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU utilization.

- Memory usage.
- Number of requests.
- Response time.
- Error rates.
- Network traffic.

When a threshold is reached, monitoring data can trigger auto-scaling.

Real-World Example

Consider an e-commerce platform such as Amazon during a major sale.

Before the sale, the platform may require 20 application instances. During the sale, traffic may suddenly increase, so auto-scaling can increase capacity to 100 or more instances. After the sale, capacity can automatically scale down.

Conclusion

Load balancing + auto-scaling + monitoring allows e-commerce applications to handle unpredictable traffic efficiently while reducing downtime and improving customer experience.

2. Secure Cloud Storage for Large Backup Data

Problem Statement

An organization needs to store large volumes of backup data securely. The storage system should provide durability, redundancy, security, and cost efficiency.

Cloud Storage and Durability

Cloud providers store data using distributed storage infrastructure. Data can be replicated across multiple storage devices and, depending on the service configuration, across different availability zones or regions.

Therefore, failure of one physical storage device does not necessarily result in data loss.

Data Redundancy

A typical architecture can be:

Primary Data

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Cloud Backup Storage

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Copy 1 | Copy 2 | Copy 3

This provides redundancy and improves data durability.

Encryption

Backup data should be encrypted:

Data at Rest: Encryption protects stored backup files.

Data in Transit: TLS/secure connections protect data while it is being uploaded or transferred.

Encryption keys should be managed securely using a key-management service.

Access Control

Access should follow least privilege.

For example:

- Backup administrator → Can create and restore backups.
- Normal employee → Cannot access backup storage.
- Auditor → Read-only access.

Additional security mechanisms include:

- IAM policies.
- Multi-factor authentication.
- Role-based access control.
- Storage access policies.
- Audit logs.

Cost Advantages

Cloud storage can be more economical because:

1. No need to purchase large physical storage infrastructure.
2. No datacenter maintenance costs.
3. Pay only for the storage consumed.
4. Storage tiers can be selected according to access frequency.
5. Backup data can be moved to lower-cost archival storage.

Disaster Recovery Example

Suppose a company's main database is accidentally deleted because of an administrator error.

The organization can:

Production Database



Automated Cloud Backup



Database Failure/Deletion



Restore Backup



Application Recovery

This allows business operations to resume without rebuilding the database from scratch.

Conclusion

Cloud storage provides durability, redundancy, encryption, access control, and flexible pricing, making it suitable for large-scale backup and disaster recovery.

3. CDN for Global Mobile Applications

Problem Statement

A mobile application company wants to deploy its application globally. Users are located in different geographical regions, so the company needs to reduce latency and improve application performance.

What is a CDN?

A Content Delivery Network (CDN) is a geographically distributed network of servers that delivers content from locations closer to users.

Traditional Architecture

User in India



Main Server in USA

This may result in higher network latency.

CDN Architecture

User in India



Nearby Edge Server



Content delivered quickly

Similarly:

User in Europe → European Edge Server

User in USA → US Edge Server

How CDN Reduces Latency

A CDN caches frequently requested content such as:

- Images.
- Videos.
- JavaScript/CSS files.
- Game assets.
- Software files.

Instead of repeatedly retrieving content from the origin server, the CDN can serve cached content from an edge location.

Role of Edge Servers

Edge servers are positioned closer to users geographically.

This results in:

- Lower network latency.
- Faster content delivery.
- Faster application response.
- Reduced buffering.
- Reduced load on origin servers.

Global Cloud Availability

Major cloud providers offer globally distributed regions and availability zones. Applications can be deployed across multiple regions to improve availability.

A global architecture could be:

Users Worldwide



Global DNS / CDN



Asia Edge | Europe Edge | US Edge



Cloud Application

Real-Time Streaming Example

Consider a live sports streaming application.

If thousands of users in Chennai request the same live video, the CDN can deliver the stream through nearby edge infrastructure rather than sending every request to a distant origin server.

This reduces latency and helps provide smoother playback.

Gaming Example

For online gaming, edge infrastructure can reduce the distance between players and game services, helping improve responsiveness and reducing network delay.

Conclusion

CDNs and edge servers improve global application performance by delivering content closer to users. Combined with multi-region cloud deployment, they provide low latency, scalability, and high availability.

4. Cloud Security and Compliance for a Financial Institution

Problem Statement

A financial institution handles sensitive information such as account details and financial transactions. It requires strict security, regulatory compliance, auditing, and data sovereignty.

Cloud Compliance

Cloud providers offer compliance-related services and certifications that help organizations meet regulatory requirements. However, the organization is still responsible for configuring its workloads correctly.

A shared responsibility model is generally used.

Role of Logging

Logging records important events such as:

- User login.
- Data access.
- Transaction activity.
- Configuration changes.
- Failed authentication attempts.
- Administrative operations.

Example:

User → Login → Transaction → Database Access

These events can be recorded in audit logs.

Role of Monitoring

Monitoring detects unusual activities such as:

- Abnormal login attempts.
- Unexpected traffic.
- High error rates.
- Unauthorized access patterns.
- Suspicious API calls.

Alerts can be generated when predefined security conditions are detected.

Role of Encryption

Financial data should be encrypted:

Data at Rest

Database → Encryption → Protected Data

Data in Transit

Client → TLS → Cloud Application

Encryption reduces the risk of sensitive data being exposed if unauthorized parties gain access to storage or network traffic.

Access Control

Financial institutions should use:

- Role-Based Access Control.
- Least privilege.
- Multi-factor authentication.
- Strong identity management.
- Privileged-access controls.

Data Sovereignty

Data sovereignty means data may need to remain within a particular country or geographical region due to legal or regulatory requirements.

The organization can:

1. Select appropriate cloud regions.
2. Restrict storage locations using policies.
3. Control where backups are stored.
4. Monitor data movement.
5. Ensure disaster recovery copies comply with location requirements.

Example – Financial Transaction

Suppose a customer transfers ₹50,000 through a banking application.

A secure flow would be:

Customer



MFA Authentication



Encrypted Connection



Transaction Service



Encrypted Database



Audit Log

The transaction is protected through authentication, encryption, access control, monitoring, and auditing.

Conclusion

Financial institutions can strengthen cloud security using encryption, IAM, MFA, logging, monitoring, auditing, compliance controls, and regional data restrictions. These mechanisms help protect sensitive financial transactions and satisfy regulatory requirements.

5. Containers vs Virtual Machines and Kubernetes

Problem Statement

A development team wants to use containerization for application deployment. They need to understand how containers differ from VMs and how orchestration platforms such as Kubernetes simplify deployment and scaling.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Virtualization	OS-level virtualization	Hardware-level virtualization
OS	Share host OS kernel	Each VM has its own guest OS
Resource usage	Low	Higher
Startup	Very fast	Relatively slower
Size	Usually smaller	Usually larger
Performance overhead	Low	Higher
Portability	High	High, but heavier
Isolation	Process-level isolation	Stronger OS-level isolation

Container Architecture

Physical/Cloud Server



Host Operating System



Container Runtime



Container 1 | Container 2 | Container 3

Containers share the host OS kernel, which makes them lightweight and efficient.

Virtual Machine Architecture

Physical Server



Hypervisor



VM 1 | VM 2 | VM 3

Each VM generally contains its own guest operating system.

Benefits of Containers

- Fast startup.
- Lower resource consumption.

- Easy deployment.
- Consistent environments.
- Easy scaling.
- Good support for microservices.

Role of Kubernetes

Kubernetes is a container orchestration platform that automates the management of containerized applications.

It can manage:

- Container deployment.
- Scheduling.
- Service discovery.
- Load balancing.
- Scaling.
- Health monitoring.
- Self-healing.
- Rolling updates.

Deployment

Instead of manually starting hundreds of containers, Kubernetes can automatically deploy them across available nodes.

Scaling

If application traffic increases:

High traffic



Kubernetes detects increased demand



More container instances



Improved application capacity

When traffic decreases, unnecessary containers can be reduced.

Self-Healing

If a container crashes:

Container Failure



Kubernetes detects failure



New container started

This improves application availability.

Portability Across Cloud Environments

The same container image can run in different environments:

Developer Laptop

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Docker

↓

Cloud Provider A

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Cloud Provider B

Because the application and dependencies are packaged inside the container, deployment becomes more consistent across environments.

Microservices Example

Consider an online food delivery application.

Instead of building one large application, it can be divided into separate microservices:

Food Delivery Application

- ├── User Service
- ├── Restaurant Service
- ├── Menu Service
- ├── Order Service
- ├── Payment Service
- └── Delivery Tracking Service

Each service can run inside its own container.

For example:

Payment Service



Container 1 | Container 2 | Container 3

Kubernetes can automatically scale the Payment Service when transaction volume increases without necessarily scaling every other service.

Conclusion

Containers are lightweight and resource-efficient compared with VMs. Kubernetes simplifies deployment, service discovery, scaling, load balancing, and failure recovery. Containerization also provides strong portability, allowing applications to run consistently across different cloud environments.